Column-Aware Reading Order for Spoken PDFs

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Abstract. We describe a reader that speaks two-column papers in the order a person would read them and removes line-end hyphens without breaking real compounds such as the phrase two-column layout. Footnote markers and citation brackets are skipped so that every word is pronounced correctly.

1 Introduction

Reading aloud scientific papers is hard because text is laid out in two columns and words are frequently hyphenated at the end of a line [3]. A naive reader follows the content stream order and jumps between the columns.[1]

We group lines into columns first, as shown in Fig. 2, and only then build paragraphs. Hyphenated words such as state-of-the-art keep their inner hyphens, while line-end hyphens are removed so the voice says each word correctly and the highlighted word stays on the page.

2 Method

Each page is cut into horizontal slabs. Slabs that share a vertical gutter are merged again, which avoids the classic XY-cut mistake.

Columns are then read top to bottom, left to right. Footnote markers and citation brackets are skipped, and abbreviations like Eq. 4 are expanded before the text is sent to the speech synthesizer.

[1] A footnote that should be read after the body.

thesizer. The result is one paragraph per reading unit, with sentences spoken one at a time.

3 Results

The reading order was correct for all pages. We used the e-mail corpus from Smith et al. and the pre-processing steps [4, 7].

Future work will handle tables and figures that span both columns of the page, because a figure

Figure 3: A caption placed inside the paragraph.

interrupts the text flow. +e same holds for tables.

$$x = a + b \tag{1}$$

4.2.1 Scalability

The tool can support $\frac{\text{5 minutes}}{\text{34 seconds}}$ = 8.8 times more jobs.

1. Each region has sufficient storage space for the tables it stores (Expression 4).
2. Short item.

The load is defined as follows:

$$L = a + b \quad (12)$$

where a is the arrival rate.

Measurements were taken every hour and

Figure 5: Load over time.

averaged over two weeks.

Prior systems [7, 8, 12] read in stream order.

Table 1: Results per voice.

Voice	Errors
Samantha	3
Ava (online)	0

The online voice made no errors [3,2,1,44,55].

References

[1] AWS Regions, 2023. https://aws.amazon.com/about-aws/global-infrastructure/.

[2] Kay Ousterhout and Ion Stoica. Sparrow: Distributed, Low Latency Scheduling. In SOSP, 2013.

Appendix

Two more tables follow.

Table 2: Design space.

		Queue scope	
		Local	Global
Resource scope	Local	(1) Borg, Hydra	(2)
	Global	(3) none	(4) Ideal

The table has headers. A grid is rebuilt from the ruling lines, so a label spanning two rows and a cell written on two lines are both read once, in the right place.

Table 3: Related work.

Work	Comment
Smith et al.	Covers columns but ignores hyphens.
Lee [4]	Reads tables row by row without headers.

Both tables are read with their column names. Without vertical rules, columns are found from the white space between them, and each wrapped cell is given to the nearest row label.